FLATTENING A FINITE MODULE ALONG AN AFFINE BLOWUP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Flatness of a finite module over a ring detected through the Fitting ideal of a presentation. the criterion counted by the sibling card S1-algebra-fitting-ideal at 1355-1411. Supplies the form in which flatness is verified at the end of the proof.
- Affine blowup of a ring along an ideal, given by adjoining the ratios of the generators. used at proof lines 6097-6130. Provides the new ring over which the module is shown to be flat.
- The equations of the blowup and their relation to the original presentation. used at proof lines 6130-6171. Supplies the columns added to the presentation matrix and the computation that follows.

Source reference. Original lemma 053I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6606–6606. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0561: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6694–6694. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0537: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6595–6595. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0539: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6597–6597. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 053E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6602–6602. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 052M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6575–6575. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02JO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3343–3343. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0538: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6596–6596. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $(R, \mathfrak{m})$ be a local domain with fraction field K. Let S be a finite type R-algebra. Let M be a finite S-module. For every valuation ring $A \subset K$ dominating R there exists an ideal $I \subset \mathfrak{m}$ and a nonzero element $a \in I$ such that*

1. *I is finitely generated,*
2. *A has center on $R[\frac{I}{a}]$,*
3. *the fibre ring of $R \to R[\frac{I}{a}]$ at $\mathfrak{m}$ is not zero, and*
4. *the strict transform $S_{I,a}$ of S along $R \to R[\frac{I}{a}]$ is flat and of finite presentation over R, and the strict transform $M_{I,a}$ of M along $R \to R[\frac{I}{a}]$ is flat over R and finitely presented over $S_{I,a}$.*

Proof. Write $S = R[x_1, \ldots, x_n]/J$ and denote $N = S \oplus M$ viewed as a module over $P = R[x_1, \ldots, x_n]$. If we can prove the lemma in case S is a polynomial algebra over R, then we can find I, a satisfying (1), (2), (3) such that the strict transform $N_{I,a}$ of N along $R \to R[\frac{I}{a}]$ is flat over R and finitely presented as a module over the strict transform $P_{I,a}$ of P. Since $P_{I,a} = R[\frac{I}{a}][x_1, \ldots, x_n]$ (small detail omitted) we find that the summand $S_{I,a} \subset N_{I,a}$ is flat over R and finitely presented as a module over $R[\frac{I}{a}][x_1, \ldots, x_n]$. Hence $S_{I,a}$ is finitely presented as an $R[\frac{I}{a}]$-algebra. Moreover, the summand $M_{I,a} \subset N_{I,a}$ is flat over R and finitely presented as a module over $P_{I,a}$ hence also finitely presented as a module over $S_{I,a}$, see Algebra, Lemma 0561. This reduces us to the case discussed in the next paragraph.

Assume $S = R[x_1, \ldots, x_n]$. Choose a presentation

$$0 \to K \to S^{\oplus r} \to M \to 0.$$

Let M_A be the quotient of $M \otimes_R A$ by its torsion submodule, see Lemma 0537. Then M_A is a finite module over $S_A = A[x_1, \ldots, x_n]$. By Lemma 0539 we see that M_A is flat over A. By Lemma 053E we see that M_A is finitely presented. Hence there exist finitely many elements $k_1, \ldots, k_t \in S_A^{\oplus r}$ which generate the kernel of the presentation $S_A^{\oplus r} \to M_A$ as an S_A-module. For any choice of $a \in I \subset \mathfrak{m}$ satisfying (1), (2), and (3) we denote $M_{I,a}$ the strict transform of M along $R \to R[\frac{I}{a}]$. It is

a finite module over $S_{I,a} = R[\frac{I}{a}][x_1, \ldots, x_n]$. By Algebra, Lemma 052M we have $A = \operatorname{colim}_{I,a} R[\frac{I}{a}]$. This implies that $S_A = \operatorname{colim} S_{I,a}$ and

$$\operatorname{colim} M \otimes_R R[\tfrac{I}{a}] = M \otimes_R A$$

Choose I, a and lifts $k_1, \ldots, k_t \in S_{I,a}^{\oplus r}$. Since M_A is the quotient of $M \otimes_R A$ by torsion, we see that the images of $k_1, \ldots, k_t$ in $M \otimes_R A$ are annihilated by a nonzero element $\alpha \in A$. After replacing I, a by a different pair (recall that the colimit is filtered), we may assume $\alpha = x/a^n$ for some $x \in I^n$ nonzero. Then we find that $xk_1, \ldots, xk_t$ map to zero in $M \otimes_R A$. Hence after replacing I, a by a different pair we may assume $xk_1, \ldots, xk_t$ map to zero in $M \otimes_R R[\frac{I}{a}]$ for some nonzero $x \in R$. Then finally replacing I, a by xI, xa we find that we may assume $k_1, \ldots, k_t$ map to a-power torsion elements of $M \otimes_R R[\frac{I}{a}]$. For any such pair (I, a) we set

$$M'_{I,a} = S_{I,a}^{\oplus r} / \sum S_{I,a} k_j.$$

Since $M_A = S_A^{\oplus r} / \sum S_A k_j$ we see that $M_A = \operatorname{colim}_{I,a} M'_{I,a}$. At this point we finally apply Algebra, Lemma 02JO (3) to conclude that $M'_{I,a}$ is flat for some pair (I, a) as above. This lemma does not apply a priori to the system of strict transforms

$$M_{I,a} = (M \otimes_R R[\tfrac{I}{a}])/a\text{-power torsion}$$

as the transition maps may not satisfy the assumptions of the lemma. But now, since flatness implies torsion free (Lemma 0538) and since $M_{I,a}$ is the quotient of $M'_{I,a}$ (because we arranged it so the elements $k_1, \ldots, k_t$ map to zero in $M_{I,a}$) by the a-power torsion submodule we also conclude that $M'_{I,a} = M_{I,a}$ for such a pair and we win. □